4. Fizik xossalari

- Arenlar – **suyuq** (oddiy sharoitda ba’zilari qattiq: naftalin) o‘ziga xos o‘tkir hidli moddalar.
- Suvda **erimaydi**, suvdan yengil (benzol zichligi 0,88 g/sm^3), organik erituvchilarda yaxshi eriydi, o‘zi ham yaxshi erituvchi.
- Molekulyar massa ortishi bilan **qaynash harorati ortadi**.
- Izomerlarda: radikallar bir-biriga yaqin bo‘lsa, qaynash harorati yuqoriroq (o- > m- > p-). **Simmetrik** tuzilgan izomerning suyuqlanish harorati yuqori (p-ksilol qotadi).

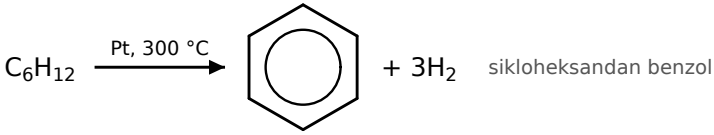
Modda	Formula	Agregat holati (20 °C)	t_{suyuq}, °C	$t_{qaynash}$, °C
benzol	C_6H_6	suyuq	5,5	80,1
toluol	$C_6H_5CH_3$	suyuq	–95	110,6
etilbenzol	$C_6H_5C_2H_5$	suyuq	–95	136
o-ksilol	$C_6H_4(CH_3)_2$	suyuq	–25	144
m-ksilol	$C_6H_4(CH_3)_2$	suyuq	–48	139
p-ksilol	$C_6H_4(CH_3)_2$	suyuq	13	138
stirol	$C_6H_5CH{=}CH_2$	suyuq	–31	145
naftalin	$C_{10}H_8$	qattiq	80	218

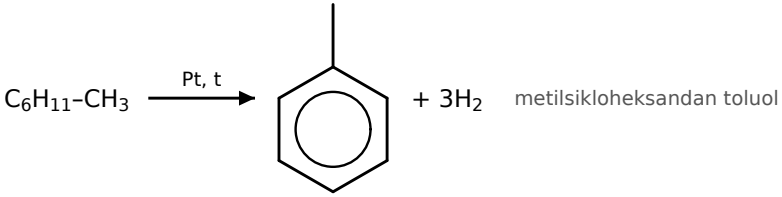
Ehtiyot bo‘ling: benzol va uning bug‘lari **zaharli** (qonga ta’sir qiladi, saraton kasalligini keltirib chiqarishi mumkin). Arenlar bilan faqat shamollatiladigan joyda ishlanadi.

5. Olinishi

Arenlar sanoatda **koks gazi** (toshko‘mir smolasi) va **neft** mahsulotlaridan ajratib olinadi, shuningdek quyidagi usullarda sintez qilinadi.

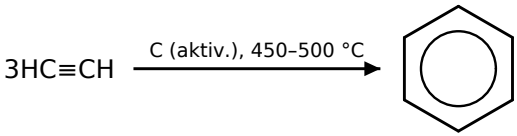
1) Sikloalkanlarni degidrogenlash (N.D. Zelinskiy)





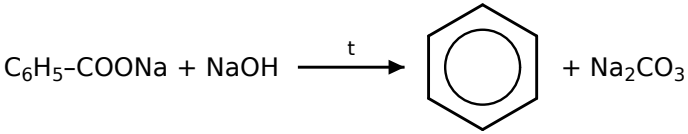
metilsikloheksandan toluol

2) Atsetilenni trimerlash (N.D. Zelinskiy)



1927-yil

3) Aromatik kislota tuzlarini NaOH bilan suyuqlantirish (dekarboksillash)



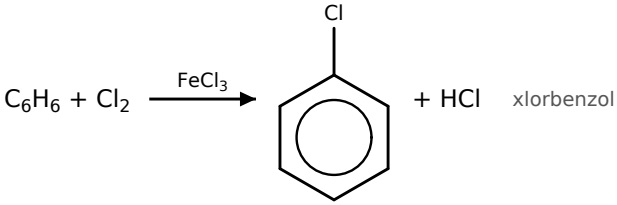
Laboratoriyada benzol natriy benzoatni NaOH bilan suyuqlantirib olinadi; **sanoatda** asosan neftni riformingi (alkan va sikloalkanlardan arenlar hosil bo‘lishi) va koks gazi manbalaridan.

6. Kimyoviy xossalari

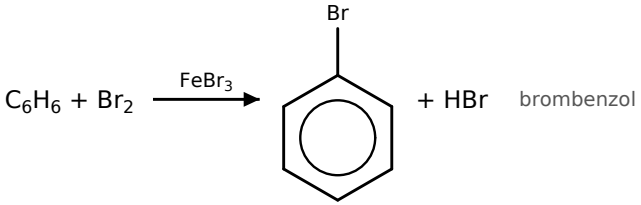
Benzolda uchta qo‘sh bog‘ borga o‘xshasa-da, aromatik sistema barqaror bo‘lgani uchun arenlar uchun birikish emas, **o‘rin olish** reaksiyalari xos. Birikish va oksidlanish faqat qattiq sharoitda boradi. Benzol gomologlarida qo‘shimcha ravishda **yon zanjir** reaksiyalari ham bo‘ladi.

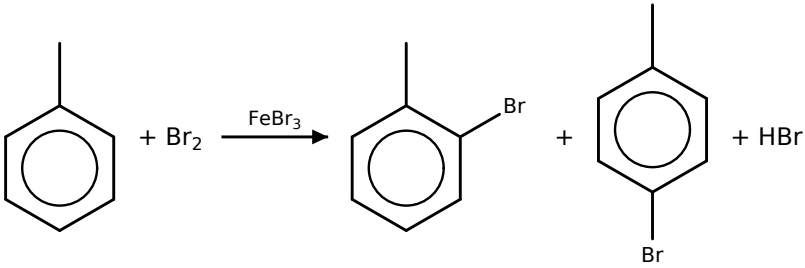
I. O‘rin olish (elektrofil) reaksiyalari

1) Galogenlash (katalizator: $FeCl_3$, $FeBr_3$, $AlCl_3$)



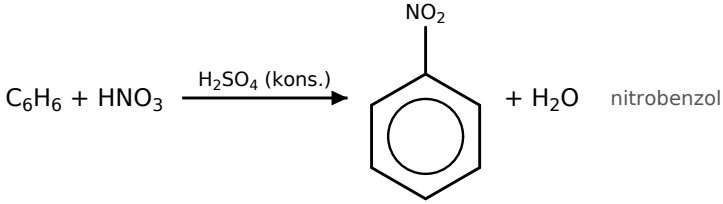
xlorbenzol

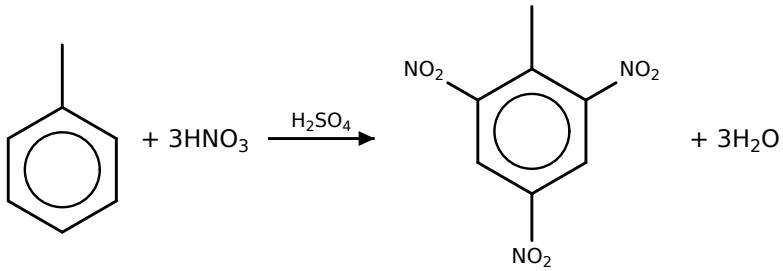




Toluolda o-bromtoluol va p-bromtoluol hosil bo‘ladi. **Benzol bromli suvni rangsizlantirmaydi**: bromli suv bilan reaksiya katalizatorsiz bormaydi.

2) Nitrolash (HNO_3 + H_2SO_4 aralashmasi)



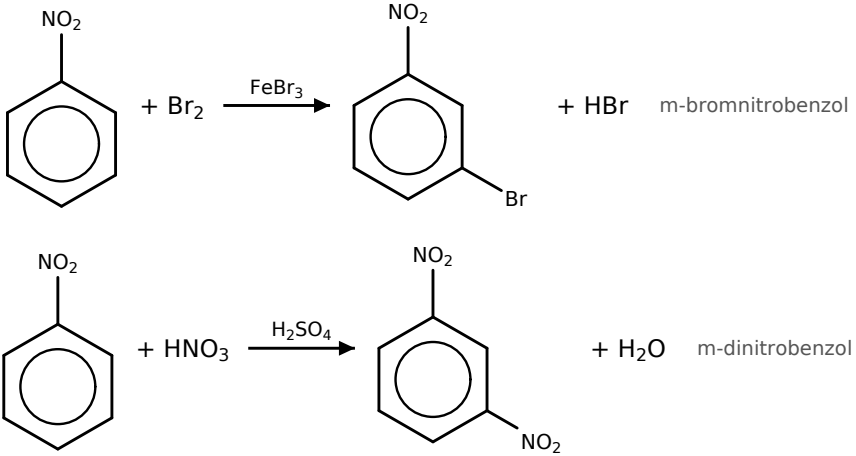


2,4,6-Trinitrotoluol (**trotil, TNT**) – portlovchi modda. Metil guruhi benzol halqasini faollashtiradi, shuning uchun toluol benzolga nisbatan osonroq (uchta H almashinadi) nitrolanadi.

Orientatsiya (yo‘naltirish) qoidasi

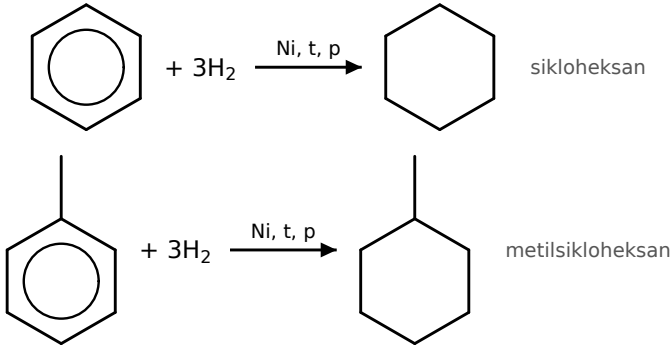
Benzol halqasida bo‘lgan o‘rinbosar **ikkinchi** kelayotgan guruhni halqaning ma’lum holatiga yo‘naltiradi:

Tur	O‘rinbosarlar	Yo‘naltiradi	Ta’siri
I tur (elektrodonor)	$-CH_3$, –R, –OH, $-NH_2$, –Cl, –Br	orto- va para-	halqani faollashtiradi (galogenlardan tashqari)
II tur (elektroakseptor)	$-NO_2$, $-SO_3H$, –COOH, –CHO, –CN	meta-	halqani sustlashtiradi

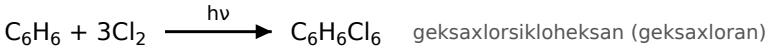


II. Birikish reaksiyalari

1) Gidrogenlash (Ni, Pt; t, p)



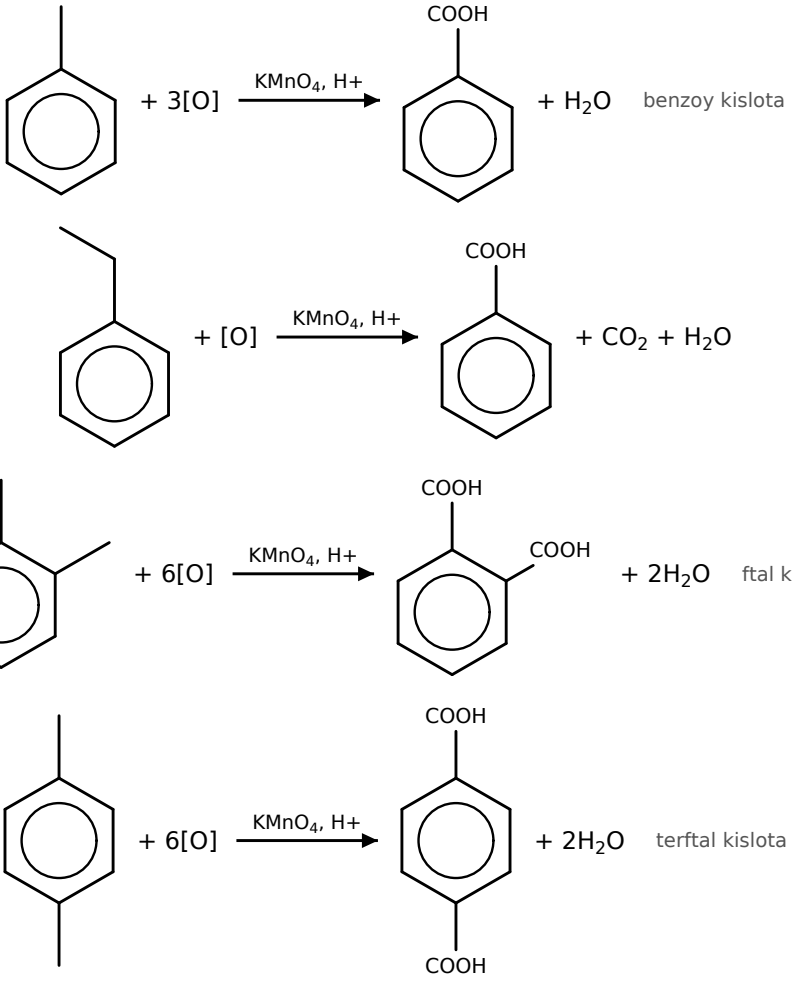
2) Xlorlash (ultrabinafsha nur)



III. Oksidlanish reaksiyalari

1) Benzol halqasi – barqaror, yon zanjir – oksidlanadi

Benzol $KMnO_4$ ta'sirida **oksidlanmaydi** (eritma rangsizlanmaydi). Benzol gomologlarida esa yon zanjir (uzunligidan qat'i nazar) **–COOH** guruhigacha oksidlanadi; halqaga birikkan uglerod –COOH ni, qolgan uglerodlar CO_2 ni hosil qiladi:



Eslatma: halqaga birikkan uglerodda vodorod bo‘lmasa (masalan, **uchlamchi butilbenzol** $C_6H_5–C(CH_3)_3$) yon zanjir oksidlanmaydi.

2) Yonish

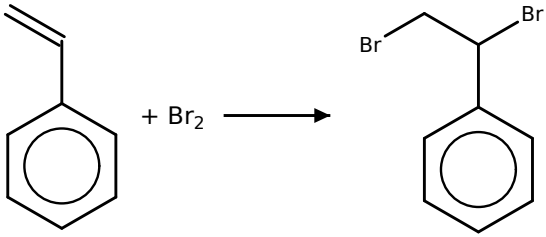
Arenlar **tutab, yorug‘ alanga** bilan yonadi (uglerodning massa ulushi katta: benzolda 92,3 %):

$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$

$C_nH_{2n-6} + (1{,}5n - 1{,}5)O_2 \rightarrow nCO_2 + (n - 3)H_2O$

IV. Stirolning xossalari

Stirolda yon zanjirda **qo‘sh bog‘** bor, shuning uchun u alkenlarga xos reaksiyalarga kirishadi va **bromli suvni, $KMnO_4$ ni rangsizlantiradi** (benzol va toluoldan farqi):



1,2-dibromfeniletan

$nC_6H_5–CH=CH_2 \xrightarrow{\text{polim.}} (–CH_2–CH(C_6H_5)–)n$ polistirol

Polistirol – shaffof plastmassa (idish, ko‘pik, izolyatsiya materiali).

Sifat reaksiyalar

Modda	Bromli suv	$KMnO_4$ eritmasi
benzol	rangsizlanmaydi	rangsizlanmaydi
toluol	rangsizlanmaydi	rangsizlanadi (isitilganda)
stirol	rangsizlanadi	rangsizlanadi

7. Ishlatilishi

Arenlar – organik sintezning eng muhim xomashyosi.

- **Benzol:** stirol, fenol, anilin, siklogeksan, bo‘yoqlar, dorilar, plastmassa va sintetik tolalar olishda; erituvchi.
- **Toluol:** erituvchi, trotil (TNT), benzoy kislota, bo‘yoq va dorilar sintezi.
- **Ksilollar:** erituvchi; terftal kislotadan lavsan tolasi olinadi, ftal kislotadan – bo‘yoq va plastifikatorlar.
- **Stirol:** polistirol va sintetik kauchuklar (butadiyen-stirol, SKS).

- **Kumol:** fenol va atseton ishlab chiqarishda (kumol usuli).
- **Geksaxloran** ($C_6H_6Cl_6$): hasharotlarga qarshi (insektitsid) vosita sifatida ishlatilgan, hozir cheklangan.
- Arenlar benzinning **oktan sonini oshirish** uchun qo‘shimcha sifatida ham ishlatiladi.

Mahsulot	Olinishi	Ishlatilishi
etilbenzol	$C_6H_6 + C_2H_4$ ($AlCl_3$)	stirol olish
stirol	etilbenzolni degidrogenlash	polistirol, kauchuk
trotil	toluol + $3HNO_3$ (H_2SO_4)	portlovchi modda
benzoy kislota	toluolni oksidlash	konservant, sintez
terftal kislota	p-ksilolni oksidlash	lavsan olish
kumol	C_6H_6 + propen	fenol, atseton

Ehtiyot bo‘ling: benzol zaharli va yonuvchan, trotil portlovchi. Ularni faqat mutaxassislar ishlatadi.

O‘z-o‘zini tekshirish

1. Benzol molekulasidagi uglerodlarning gibridlanishi va valent burchagini yozing.

2. Nima uchun benzol bromli suvni rangsizlantirmaydi, stirol esa rangsizlantiradi?

3. C_8H_{10} va C_9H_{12} tarkibli arenlar nechta izomerga ega?

4. o-ksilolni $KMnO_4 + H_2SO_4$ bilan oksidlash mahsulotini yozing.

5. Toluolga $FeCl_3$ ishtirokida xlor ta’sir ettirilganda qanday moddalar hosil bo‘ladi?

6. 7,8 g benzol to‘liq yonishi uchun necha litr (n.sh.) kislorod kerak?

7. 15,6 g benzoldan nitrolash yo‘li bilan necha gramm nitrobenzol olish mumkin (unum 100 %)?

8. 44,8 l (n.sh.) atsetilen trimerlanganda necha gramm benzol olinadi?

9. Vodorod bo‘yicha zichligi 53 bo‘lgan arenning formulasini toping va izomerlari sonini yozing.

10. 16,8 g sikloheksandan unum 80 % bo‘lganda necha gramm benzol olinadi?

Javoblar: 1) sp^2, 120°; 2) benzolda qo‘sh bog‘ yo‘q (π-elektronlar delokalizatsiyalangan), stirolning yon zanjirida oddiy C=C bor; 3) 4 ta va 8 ta; 4) ftal kislota (benzol-1,2-dikarbon kislota); 5) o-xlortoluol va p-xlortoluol; 6) $n(C_6H_6) = 0,1$ mol, $n(O_2) = 0,75$ mol, $V = 16,8$ l; 7) $n = 0,2$ mol, $M(C_6H_5NO_2) = 123$, $m = 24,6$ g; 8) $n(C_2H_2) = 2$ mol, $n(C_6H_6) = 2/3$ mol, $m = 52$ g; 9) $M = 106$, $14n – 6 = 106$, $n = 8$, C_8H_{10}, izomerlari 4 ta; 10) $n(C_6H_{12}) = 0,2$ mol, $0,2 \cdot 78 = 15,6$ g, $15,6 \cdot 0,8 = 12,48$ g.